

F1.15 Ornament as a global language; the exhibition as taxonomy

Patterns are related by how they were made, by who met whom, or by who displayed them together, and the museum record shows which.

Spec

Row	F1.15
Essential question	When two patterns from different worlds look alike, who decided they belong together: the makers, or the people who printed, exhibited and catalogued them?
Opening move	Prediction first. Six unlabelled patterns appear on the wall: a Mamluk inlaid bowl, a Kuba raffia pile cloth, a Tongan cloth pattern from Owen Jones's plates, an Inka khipu, a girih star panel (our own drawing) and a Yingzao fashi bracket module (our own drawing). The learner links the ones they predict are "related" and says why. The reveal sorts every link into three kinds: shared method, documented contact, or shared label given by a book or exhibition. Claim: "All ornament should be based upon a geometrical construction" (Jones, 1856, Proposition 8). Counter-claim: "evolution der kultur ist gleichbedeutend mit dem entfernen des ornamentes aus dem gebrauchsgegenstande" (Loos, lecture 1910). Case: the Cleveland bowl 1945.133, published in 1946 as a "Venetian" bowl and now catalogued as Damascus, 1450–1500.
Lab or main interactive	Pattern Comparer. The learner sets any two patterns side by side, predicts "related / not related", then opens three layers: the making rule (a construction diagram), the route (documented contact, with dates), and the label history (every book, exhibition and catalogue heading that grouped them, pulled from the museum record's exhibition field). Keyboard-operable; each pattern has alt text that states its rule, not its look.

Row	F1.15
Concept	About 900 words. (1) 1851 and 1856: London sorted the world's goods by country and class; Jones sorted its ornament by people and period, beginning with "Savage Tribes". (2) The counter-position: Loos makes the removal of ornament the measure of culture, using a "Papuan" as his foil. (3) Geometry as method: ornament as a making system with its own theory (giri, the Topkapı Scroll, muqarnas, the Yingzao fashi's cai-fen module, the khipu's cord grammar, Vanuatu sand drawing's continuous line). (4) The exhibition as taxonomy: every record's exhibition history is data; the Atlas reads it to show when objects were grouped, renamed or ranked.
Cases	Three object stories. A. The bowl that was "Venetian" (Cleveland 1945.133): Mamluk inlayers in Damascus, a 1946 attribution to Venice, a 1956 "Venetian Tradition" show, a 2006 geometry show. B. The designer paid in cloth (AS-001, Jones 1856): a dress brought from Tongatapu by Oswald Brierly, and Jones's note that one woman designed all the island's patterns. C. The raffia pile cloth (Cleveland 1991.224): Kuba men weave, women embroider; the pattern system and its later reading by mathematicians and curators.
Apply — maker	In the Pattern Comparer, build one pattern from a declared method (compass grid, module, or continuous line) and write the rule into GRM.f in one sentence another maker could follow without seeing the image.
Apply — briefer	For one reference pattern in your brief, name what was extracted and from whom: the maker community (own name first), the channel (book, exhibition, museum, platform), the date, and one thing the brief will not take. Written into SRC.

Row	F1.15
Partner	Historian. Question: "Who drew the line between these two patterns, and when?" The test it ends with: pick one pair on your wall, say whether the relation is method, contact or label, and point to the record field or source line that proves it. If you cannot point, the Historian marks the relation "claimed, not shown".
Forms	3 (object story), 10 (comparison wall); the opening uses a prediction step inside form 10.
Consumption forms	Five-minute: the six-pattern prediction and its reveal. Session: about 50 minutes: prediction, concept, three object stories, Pattern Comparer, Apply, Historian test. Reference: the comparison wall as a standing page, each pattern with its record, rule diagram, label history and claims.
Media	Cleveland Open Access images (CC0); plates from AS-001 (1856, public domain) with the exclusion in section 2; our own construction diagrams (girih on a compass grid, cai-fen proportions, a generic continuous line on a grid); link-outs for kōwhaiwhai, Vanuatu sand drawing and the Topkapı Scroll. No generated images.
Artifact fields	GRM.f (the making rule); SRC (the extraction line).
Threads	Attribution and Authority (who is credited for a pattern, and who classified it); Writing and Argument (Jones against Loos, as positions with dates); Accessibility level 1 (alt text that states a pattern's rule; pattern contrast checked; the Comparer usable without a mouse).
Closing contribution	One SRC line: "Pattern [name] — made by [community or maker], reached me via [book / exhibition / record, date]; relation claimed: [method / contact / label]; not taken: [what]."
Claims sheet	Section 3: 20 claims, with sources opened and their depth.

1. Responsibility block

The unit must reach these canon groupings and types. Owen Jones PER415, Adolf Loos PER425, the Great Exhibition EVT001, the design reform movement MOV004 and the South Kensington Museum INS052 carry the European classifying story. Islamic geometric pattern construction GAP026, muqarnas TEC168, the Mamluk Sultanate POL042 and Mamluk style STY104 carry the geometric theory of West Asia and North Africa. The Kuba Kingdom POL022, Kuba peoples COM042, Kuba textile design STY089, Kuba raffia textile making PRA006 and technique TEC125 carry central Africa. Khipu OBT139, khipu construction TEC104 and the khipukamayuk MKR024 carry the Andes. Li Jie PER164, the Yingzao fashi TXT092 and the Song POL166 (Northern Song period PRD099) carry East Asia. Vanuatu sand drawings PRA212 with Ni-Vanuatu COM278, and Kakai Tonga COM291 with barkcloth OBT047, carry the Pacific. Māori COM288 enters only through the kōwhaiwhai link-out.

Sub-regions: WA-LEV (Damascus), WA-IRN (Isfahan), AF-CEN, AM-AND, AS-CHN, OC-MEL, OC-POL, EU-BLC, EU-WCE. Periods run from 1103 (Yingzao fashi) to 2006 (the Cleveland geometry show).

World chapters reached with a case or example: F1.10 Islamic world (case A; girih), F1.6 Africa (case C), F1.12 Australia and the Pacific (case B; sand drawing; kōwhaiwhai), F1.8 East Asia (Yingzao fashi), F1.7 Americas (khipu), F1.11 Europe and the Mediterranean (1851, Jones, Loos). That is six of nine.

The Atlas carries instead: Semper PER416 and the Trinidad hut at 1851; Riegl PER426 and Stilfragen; Birdwood PER242; Moorish Revival STY032; Samarra styles STY100; saz style STY110; zellige PRA009; the later world's fairs EVT002–EVT016; Primitivism STY062 and MoMA 1984 EVT026 (to F1.2 and F1.24); Gordon Walters PER356 (to F1.2).

2. Protocol screen

- **Excluded: the Chester head.** Jones's first chapter reproduces a tattooed head "from the museum at Chester". The Met identifies it as a preserved Māori head (mokomokai), then thought to be a woman's. That is human remains. Its woodcut and any plate showing it are removed from every use of AS-001, including thumbnails and the Pattern Comparer. The story may say that Jones used it, in words only, with toi moko repatriation linked (OCC266, OCC271).
- **Kōwhaiwhai.** Painted rafter sequences in a wharehau depict the descendants of the founding ancestor. We show no specific kōwhaiwhai image and draw none. Link out to Deidre Brown's Te Ara entry only. A Māori reader clears the wording before G3. Notice

required.

- **Vanuatu sand drawings.** UNESCO notes drawings carry "sacred or profane meanings". We reproduce no named drawing. Our diagram shows only a generic line on a grid, captioned as a diagram of the method, not a sand drawing. Notice required; the holders' own site or the UNESCO page is the image source, linked.
- **Khipu (1940.469).** Indigenous-community material with no recorded findspot. Show it; Notice on the Andean communities that kept khipu into the 20th century; say the findspot is unknown.
- **Kuba cloth (1991.224).** PRA006 notes some cloths are buried with the dead. This cloth's use is not recorded; no content note needed, but do not call it a burial cloth. Notice on the Kuba peoples.
- **Tongan cloth (case B).** Jones calls its maker's people a "Savage Tribe". The phrase appears only in quotation marks, attributed, once. Notice on Kakai Tonga.
- **Loos's "Papua".** Quoted once, attributed, with a line saying the claim is his and was used to rank peoples.
- No altar objects or grave goods are used. No person is described as goods.

3. Claims

1. *Jones's Grammar of Ornament (London: Day and Son, 1856) has 100 chromolithographic plates in its first folio edition, with contributions by J. B. Waring, J. O. Westwood and M. D. Wyatt.* Documented. https://archive.org/details/McGillLibrary-rbnc_grammar-ornament_elfNK1510J71856-22115 — catalogue record.
2. *Its 20 chapters run from "Ornament of Savage Tribes" to "Leaves and Flowers from Nature", sorting ornament by people, place, faith and period.* Documented (chapter list); the reading "sorting by people" is interpretive. https://archive.org/stream/jonesowengrammar00owen/jonesowengrammar00owen_djvu.txt — full text (OCR of an 1856 copy).
3. *Proposition 8 reads: "All ornament should be based upon a geometrical construction."* Documented. Same URL — full text.
4. *Jones reports that on Tongatapu one woman designed all the patterns in use, and was paid a number of yards of cloth for each new one; the dress came via Oswald Brierly.* Documented as Jones's report; her name is not recorded. Same URL — full text.
5. *Jones reproduced a preserved, tattooed Māori head (mokomokai) from the Chester museum, then thought to be a woman's.* Documented. <https://www.metmuseum.org/perspectives/grammar-of-ornament> (Speelberg and Fleming, 2018) — full article; Jones's footnote in the full text above.

6. *Jones decorated the 1851 building in red, yellow and blue and helped select objects from the exhibition for the School of Design's teaching collection.* Documented.
<https://www.vam.ac.uk/articles/owen-jones-and-the-grammar-of-ornament> — full article.
7. *The 1851 exhibits were shown by country and divided into four divisions and 30 classes; over six million people visited.* Documented.
<https://www.rct.uk/group/381/content/explore/themes/great-exhibition-of-1851/the-exhibition> — summary.
8. *The 1851 catalogue's introduction frames the show by "nations" and "every civilized people"; the word "race" does not appear in the introduction we read.* Documented for the text read.
https://archive.org/stream/1851OfficialDescriptiveCatalogueSection1Intro_201711/EX.1851.22_djvu.txt — full text (OCR). The brief's "nation and race" framing therefore rests on Jones's chapter order and later exhibitions, not on this text: interpretive.
9. *Loos first gave "Ornament und Verbrechen" as a lecture in 1910; it appeared in French in 1913 and in German in 1929; the 1908 date was added for the 1931 reprint.* Contested.
 Reading one: 1910, following Christopher Long's research as cited. Reading two: 1908, the date in Loos's own 1931 collection and many anthologies.
https://en.wikipedia.org/wiki/Ornament_and_Crime — summary; Long not opened.
10. *Loos wrote that the Papuan who tattoos "is no criminal", the modern man who does is "a criminal or a degenerate", and that the evolution of culture means removing ornament from everyday objects.* Documented. <https://www.designtagebuch.de/wp-content/uploads/2011/11/adolf-loos-ornament-und-verbrechen.pdf> — full text (from *Sämtliche Schriften*, 1962).
11. *Girih patterns were long read as lines drawn with compass and straightedge; Lu and Steinhardt argue that by about 1200 CE designers also used a set of five tile shapes, and that the Darb-i Imam shrine (Isfahan, 1453) is nearly quasi-crystalline.* Contested.
 Reading one: tile method, as argued. Reading two: a published comment in *Science* (2007) disputes the quasi-crystal reading.
https://www.physics.rutgers.edu/~pchandra/physics601/Lu_Steinhardt.pdf — full text; the comment was not opened.
12. *The Topkapı Scroll (Timurid, late 15th or early 16th century) holds 114 geometric patterns for wall surfaces and vaulting.* Documented.
<https://www.getty.edu/publications/virtuallibrary/9780892363353.html> — summary (free full PDF linked there, not read).

13. *The Yingzao fashi* was completed by 1100 and printed in 1103; its 34 chapters cover measurement, carpentry, stonework, tiling, and painting and decoration. Documented. https://en.wikipedia.org/wiki/Yingzao_Fashi — summary.
14. *Its timber module, the cai, has eight grades; each cai divides into 15 by 10 fen, and every component is measured in fen.* Documented. <https://www.archdaily.com/949479/from-ancient-to-modern-modular-construction-in-chinese-timber-architecture> (Miao, 2020) — summary.
15. *Vanuatu sand drawings are on UNESCO's Representative List (2008, 3.COM), first proclaimed a Masterpiece in 2003; drawers trace one continuous line on an imagined grid; UNESCO names a purely aesthetic reading as a threat.* Documented (harvested). <https://ich.unesco.org/en/RL/vanuatu-sand-drawings-00073> — full page.
16. *Kuba men weave the raffia and women embroider it; one analysis finds Kuba designs use 12 of the 17 plane symmetry types.* First part documented; the count is probable. It rests on Donald Crowe, *Journal of Geometry* 1 (1971): 169–182, which we saw only as a record. https://en.wikipedia.org/wiki/Kuba_textiles — summary.
17. *Cleveland 1945.133 was published in 1946 as a "Venetian Bronze Bowl", shown in "The Venetian Tradition" (1956–57), and is now catalogued as Damascus, Burji Mamluk, 1450–1500; its panels hold six designs, each twice.* Documented. <https://openaccess-api.clevelandart.org/api/artworks/1945.133> — full record.
18. *Exhibition histories group objects: 1945.133 and the Kuba cloth 1991.224 were both in "The Persistence of Geometry" (MOCA Cleveland, 2006); the khipu 1940.469 was in the "Exhibition of Peruvian Art" at the "Museum of Primitive Art", New York (1963–64).* Documented. <https://openaccess-api.clevelandart.org/api/artworks/1991.224> and <https://openaccess-api.clevelandart.org/api/artworks/1940.469> — full records.
19. *Six Santa Valley khipus match a colonial census record; colour and the direction of cord attachment appear to encode villagers' names.* Documented for the match; interpretive for names (as CL-166). <https://dukeupress.wordpress.com/2018/01/31/groundbreaking-study-on-incan-khipus-published-in-ethnohistory/> — summary.
20. *Unit framing: a likeness between patterns is a claim of method, contact or label, and exhibitions and catalogues make most label claims.* Interpretive (ours).

Not used as a claim: Ron Eglash's *African Fractals* (1999). We read only Bangura's review (*Nexus Network Journal*, 2000; <https://link.springer.com/content/pdf/10.1007/s00004-999-0019-3.pdf>, full text). It reports Eglash's argument that recursive scaling is intentional in African settlements, braiding and windscreens, and the reviewer's two questions: what

distinguishes African fractals, and how self-similarity is measured. Eglash may appear in the Mastery reading list as an argued position, attributed. He should not appear as a fact until the Historian has read the book.

4. Nodes

- Case A: POL042, STY104, GAP026, CIV086; exhibition node proposed below.
- Case B: PER415, TXT026, AS-001, EVT001, MOV004, INS052, COM291, OBT047, TEC108, MKR088.
- Case C: POL022, COM042, STY089, PRA006, TEC125, MAT067.
- Geometry strand: GAP026, TEC168, OBT139, TEC104, MKR024, INV005, TXT092, PER164, PER174, TEC135, POL166, PRA212, COM278.
- Counter-position: PER425, TXT030.
- Claims 1–8 → PER415, TXT026, EVT001, INS052; 9–10 → PER425, TXT030; 11–12 → GAP026, TEC168; 13–14 → TXT092, PER164, TEC135; 15 → PRA212; 16 → STY089, TEC125; 17–18 → proposed EVT rows; 19 → OBT139, TEC104.

Proposed new canon rows:

- Topkapı Scroll (Topkapı Palace Museum Library MS H.1956) — primary text (TXT).
- Kōwhaiwhai — technique / living practice, F1.12, Indigenous-community.
- "The Persistence of Geometry" (Cleveland Museum of Art at MOCA Cleveland, 2006) — event (EVT).
- "The Venetian Tradition" (Cleveland, 1956–57) — event (EVT).
- "Museum of Primitive Art" (New York) — institution (INS); dates to verify.
- Official Descriptive and Illustrated Catalogue of the Great Exhibition (1851) — primary text (TXT).
- Oswald Brierly — person (PER), as the channel for case B.
- Daniel Sheets Dye, *A Grammar of Chinese Lattice* (1937) — primary text (TXT), a second "grammar" for comparison.
- Riegl, *Stilfragen* (1893) — primary text (TXT).

5. Candidate open objects

Object	Holder, accession	Date	Licence as stated	1970 status	Record
Bowl with Geometric Designs (sheet brass inlaid with silver and gold)	Cleveland Museum of Art, 1945.133	1450–1500	CC0	n/a (decorative); Brummer Gallery, ?–1945, then Cleveland	https://clevelandart.org/art/1945.133
Embroidered Raffia Pile Cloth, Kuba peoples	Cleveland Museum of Art, 1991.224	"1900s" (1900–1990)	CC0	n/a (20th-century textile); no provenance published — flag	https://clevelandart.org/art/1991.224
Inka khipu	Cleveland Museum of Art, 1940.469	c. 1400–1532	CC0	Pass: in Cleveland since 30 Oct 1940 (WS-053); findspot unknown	https://clevelandart.org/art/1940.469
The Grammar of Ornament, first edition (AS-001)	McGill University Library copy, via Internet Archive	1856	No rights statement on the page; public domain by date	n/a	https://archive.org/details/McGillLibrary-rbbsc_grammar-ornament_elfNK1510J71856-22115
A Grammar of Chinese Lattice, vol. 1	HathiTrust (Ohio State and University of California copies)	1937	"Full view" (US); rights code not captured — to verify before use	n/a	https://catalog.hathitrust.org/Record/000566990

No open object for Vanuatu sand drawing or kōwhaiwhai: link out (UNESCO; Te Ara), plus our own generic method diagram for the sand-drawing line only. No open object for the Yingzao fashi: our own cai-fen diagram, with a link to the Chinese text.

6. People and credit

Named in the records: Owen Jones; his contributors J. B. Waring, J. O. Westwood and M. D. Wyatt; Oswald Brierly; Adolf Loos; Li Jie; Liang Sicheng; Helen Foote (1946 Cleveland bulletin); Henry Sayles Francis (1956 catalogue); Lowery Stokes Sims (2006 catalogue); Donald Crowe. Living scholars cited: Peter J. Lu, Paul J. Steinhardt, Gülru Necipoğlu, Deidre Brown, Manuel Medrano, Gary Urton, Ron Eglash, Abdul Karim Bangura.

Who the credit leaves out: the Tongan woman who designed an island's patterns, paid in cloth, unnamed by Jones. The Damascus inlayers of 1945.133, whose work was credited to Venice for at least a decade. The Kuba women who embroidered 1991.224, recorded only as "Kuba peoples". The khipukamayuy who made 1940.469. The lithographers at Day and Son who drew the stones. The builders who made girih panels from the scroll's drawings. The Song carpenters whose practice Li Jie wrote down. Ni-Vanuatu drawers, credited as a nation on the UNESCO list.

7. Visual plan (zero cost)

- **Form 3, case A:** Cleveland CC0 image of 1945.133; our own diagram of one of its six panel designs, drawn from the CC0 photograph as a construction on a compass grid.
- **Form 3, case B:** AS-001 plate with the Tongan cloth patterns (public domain), minus the excluded woodcut; our own timeline strip (1851 → 1856).
- **Form 3, case C:** Cleveland CC0 image of 1991.224; exhibition-history strip drawn from the record.
- **Form 10, comparison wall:** the three CC0 images and the khipu 1940.469 (CC0); our own diagrams: girih five-tile set, cai-fen proportions, a generic continuous line on a dot grid. Link out: Topkapı Scroll (Getty), Darb-i Imam (Wikimedia, per-item licence to check), kōwhaiwhai (Te Ara), sand drawings (UNESCO; photos © Vanuatu National Cultural Council).
- **Pattern Comparer:** the same set; label layers built from Cleveland's exhibition fields (data, no image cost).
- No generated images of historical objects, people or places.

8. Success criteria

1. In the Pattern Comparer, the learner labels at least one pair as method, contact or label, and cites the record field or source line that shows it.
2. The learner's GRM.f entry states a making rule that a second learner can follow to draw a matching unit without seeing the original.
3. The Briefer's SRC line names the maker community by its own name, the channel and date of extraction, and one thing the brief will not take.

9. Readings and risks

Historian pre-read (pack v2) should check: the Cleveland 1946 bulletin (JSTOR 25141303) for why the bowl was called Venetian, and when the label changed; the 2007 Science comment on Lu and Steinhardt, and their reply; Christopher Long on the Loos date; Crowe (1971) for the Kuba count; Necipoğlu's Topkapı Scroll (Getty PDF) for how the scroll was

used; Eglash (1999) itself, before any use; the Yingzao fashi's "lesser carpentry" chapters, to confirm lattice windows are covered before we say so; the Yuanye (TXT096) for a Chinese lattice theory of its own.

Hold until a reader clears: any kōwhaiwhai wording (Māori reader); the sand-drawing diagram and caption (a ni-Vanuatu or Vanuatu Cultural Centre reader, if reachable); the Loos quotation and its framing; the Tongan case wording (a Tongan reader).

10. Fact-check list

- R1/R2 canon rows used: EVT001, OBT139, OBT047, COM042, COM288, POL022, MOV004 (R1); PER415, GAP026, TEC104, MKR024, INS052, PRA006, STY089, TEC108, PRA204 (R2).
- PER356 spells "kowhaiwhai" without a macron; correct to kōwhaiwhai.
- TXT030: the German text on de.wikisource was not reachable; confirm it and its source edition.
- Living people: Lu, Steinhardt, Necipoğlu, Brown, Medrano, Urton, Eglash, Bangura, Sims — names, spellings, affiliations.
- Endonyms: Tongatapu and Kakai Tonga; Kuba and Bushoong; Ni-Vanuatu; Māori, kōwhaiwhai, wharenuī; Inka; Darb-i Imam.
- UNESCO: Vanuatu sand drawings, RL 2008 (3.COM), Masterpiece 2003 — confirm at source again at G3.
- Licences: Cleveland 1945.133, 1991.224, 1940.469 (CC0); AS-001 McGill scan (no stated rights); HathiTrust Dye vol. 1 rights code.
- Jones text: confirm the plate number for the Tongatapu dress (the OCR places it near Plate III, page 14) and the Chester footnote wording.
- 1851: confirm "30 classes" and the four divisions against the official catalogue, and that "race" is absent from the introduction.
- The "Museum of Primitive Art": founding and closing dates.
- Author of the 2007 Science comment: not verified.